Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
December 2025

Max. Marks: 100 | **Duration: 3Hrs.**
Class: T. Y. COMP , CSE | **Semester: V**
Course Code: CE304/ CS304 | **Date: 11/12/2025**
Subject: Theory of Computation

Instructions: (1) All questions are compulsory.
(2) Assume any necessary data but justify the same.

Course Outcomes: At the end of the course students will be able to
CE/CS304.1 Design finite automata for regular expressions and languages. CE/CS304.2 Understand and apply properties of regular languages.
CE/CS304.3 Design context-free grammar for a language and convert it into normal forms.
CE/CS304.4 Design Pushdown Automata and Turing Machine for a language.

Q No	Questions	Max Marks	CO	BL
Q.1A	Explain, with reasoning whether there is a Turing Machine that decides whether another Turing Machine ever prints the symbol 1 during its computation.	10	4	3
Q1B	Given the following TM (transitions shown), trace its moves on input 010:	10	4	3

State	Read	Write	Move	Next
q0	0	X	R	q1
q0	1	1	R	q0
q1	0	0	R	q1
q1	1	X	L	q2
q2	0	0	L	q2
q2	X	X	R	q0

Q2A	State whether each is decidable or undecidable: 1. ATM={⟨M,w⟩|M accepts w} 2. HALTTM={⟨M,w⟩|M halts on w} 3. ETM={⟨M⟩|L(M)=∅}	10	4	3
Q2B	Is the complement of a recursively enumerable language also recursively enumerable? Explain why/why not.	10	2	3
Q3A	Give a CFG to generate: L={ $a^i b^j c^k$ | i,j,k ≥ 0 and either i=j or j=k}. Is the grammar ambiguous? Why or Why not? Justify your answer.	10	3	3
Q3B	Convert the following CFG to CNF (Chomsky Normal Form): S → AB A → aAA | ε B → bBB | ε OR Consider the following grammars G1, G2, G3, G4 defined over the alphabet Σ={a,b} for each grammar Grammar 1: G1 S → aSbb | aA A → bA | ε Grammar 2: G2 S → aSB | AB A → aA | a B → Bb | b Grammar 3: G3 S A → A S b A a → a A A b → b b Grammar 4: G4 A B a → B A b A a → a A B b → b B B → ε For each grammar, 1. Identify which type of grammar it belongs to according to the Chomsky hierarchy (Type 0 – Type 3). 2. Justify your answer by explaining how the grammar satisfies the structural rules of that grammar type. 3. justify your answer whether the grammar can generate an infinite language or not.	10	3	3
Q4A	Construct a context free grammar for the language:	10	3	3

	L= {w | w ε (a, b)*, such that w is palindrome of odd length and contains equal number of a's and b's} (i) Prove that your grammar correctly generates the required language. (ii) Determine whether the string w=ababa belongs to L. (iii) Comment on whether your grammar is ambiguous or not, with justification.			
Q4B	Design a Mealy machine that outputs 1 only if the string so far is composed of alternating 0 and 1 pairs (e.g. 01, 0101, 010101, etc.). The output should be 1 whenever the sequence read up to that point is of the form $(01)^+$, i.e., one or more repetitions of "01". Convert it into a Moore machine.	10	1	3
Q5A	Construct a PDA for the given language: $L = \{ ab^n ac^m ab^n a \mid n \geq 1, m \geq 1 \}$. Show one example of acceptance and rejection by tracing through the constructed PDA. **OR** Construct an NPDA for $L=\{ a^i b^j c^k \mid i=j \text{ or } j=k, i,j,k \geq 1 \}$. Show one example of acceptance and rejection by tracing through the constructed PDA.	10	4	3
Q5B	Consider the following CFG G = (V, Σ, P, S), where V = {S, T, X}, Σ = {a, b}, the start variable is S, and the production rules P are $S \rightarrow aT\ Xb$ $T \rightarrow XT\ S \mid \varepsilon$ $X \rightarrow a \mid b$ Convert G to an equivalent PDA. Show one example of acceptance and rejection by tracing through the constructed PDA.	10	4	2

Course Outcome	Marks in MSE	Marks in ESE
CO1	15	10
CO2	15	10
CO3	-	30
CO4	-	50